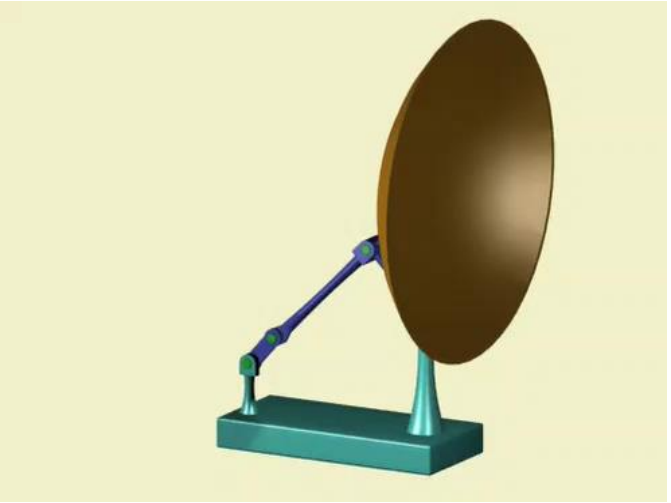
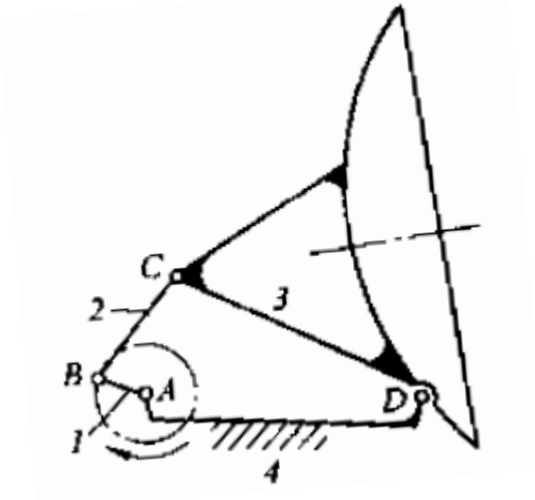
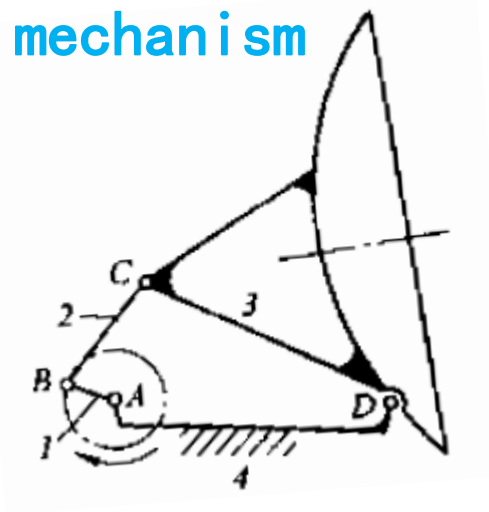
how a mechanical engineer “decompose” a machine?

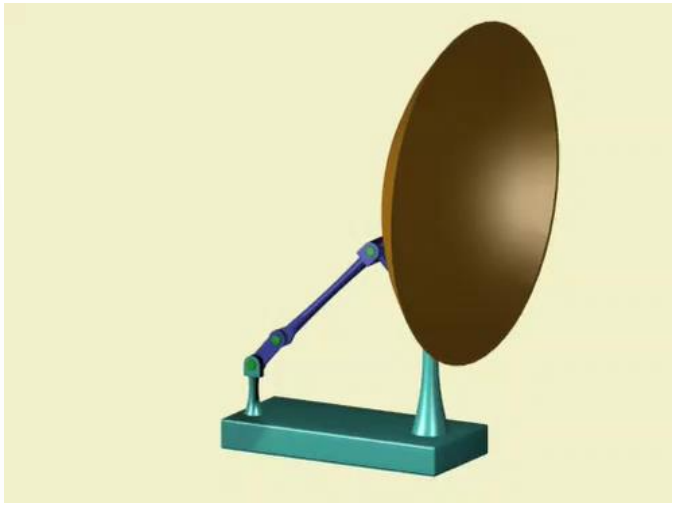
Radar mechanism





Radar mechanism





What is mechanism?

Transit or transform motion and force

Look at this radar mechanism. What can you see?

When we analyze motion, do we really care how many screws, bearings or bolts are inside?

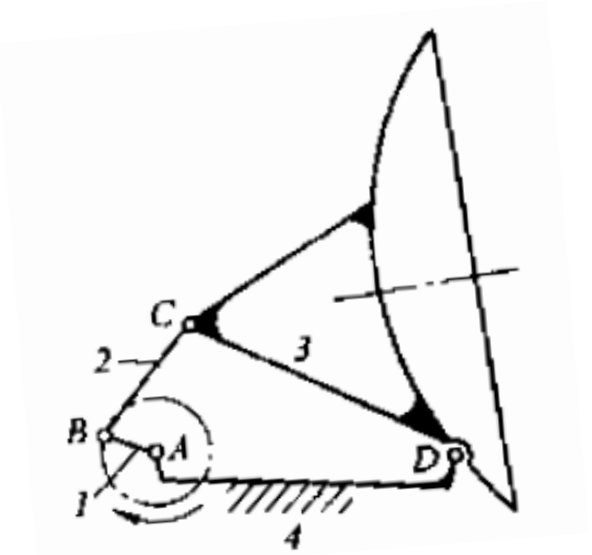
If several parts are rigidly fixed together and always move together, should we treat them as several moving bodies or one?

That rigid moving unit is called a **link**.

Is every part a link? What is the difference

Part → manufacturing
Link → motion

Categories of Links

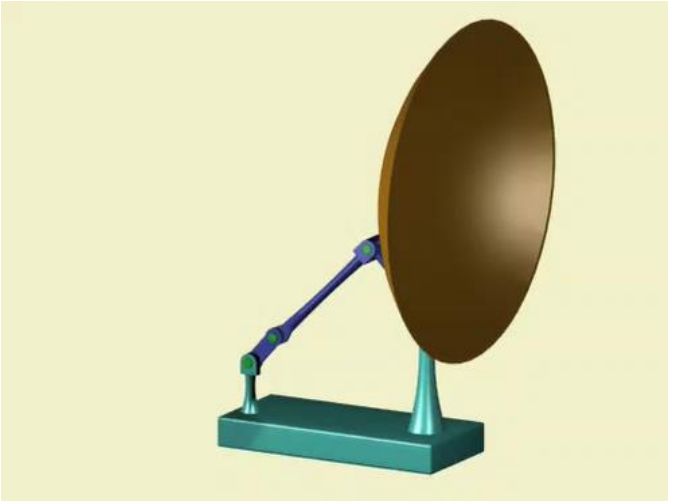


Which link never moves?

Link 4 and We call it the **frame** or **base**

Which link receives motion from outside the mechanism?

Link 1. That is the **input link**, or the **driving link**.



Which link gives us the motion that we actually want?

Link 3. That is the **output link**.

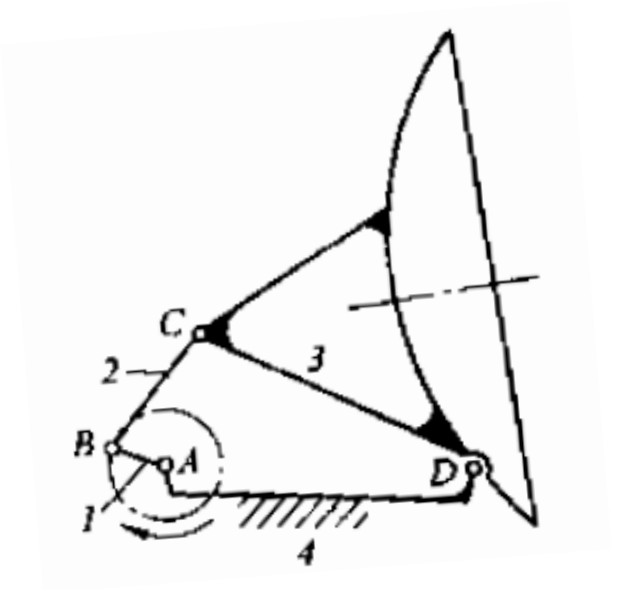
What about the links in between?

Link 2. We call them **intermediate links** or **driven links**.

We already have links.
But if I simply weld two links together, do I have a mechanism?

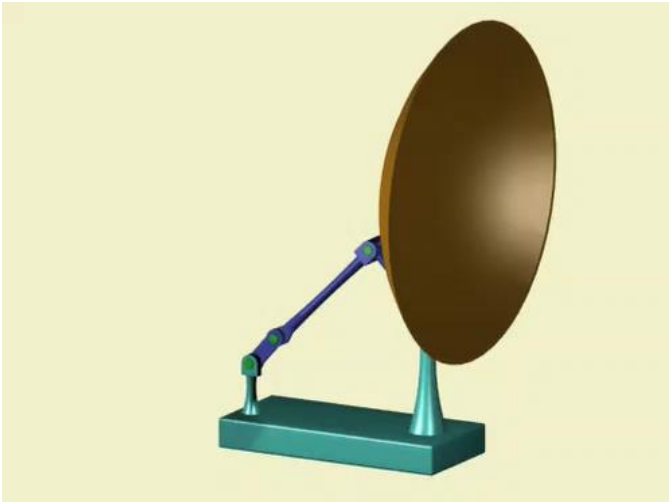
A mechanism needs links, but it also needs controlled relative motion between links.

The connection that allows a specific relative motion between two links is called a **kinematic pair**, or more commonly, a **joint**.



What does a joint actually do?

Joint = allowed motion + constrained motion

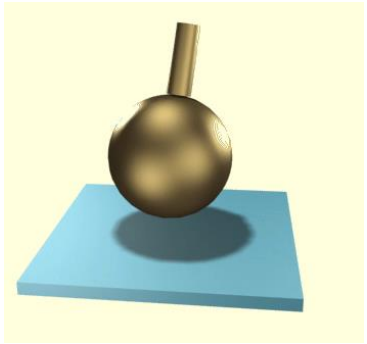


How do these two bodies contact each other?
Through a surface, a line, or a point?

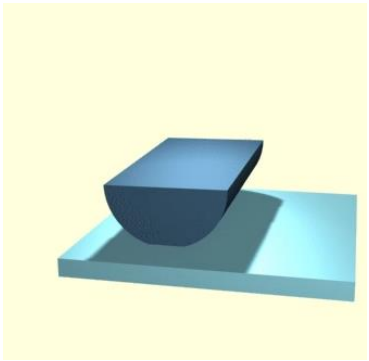
So one way to classify joints is based on their **contact geometry**

Surface contact → lower pair
Point or line contact → higher pair

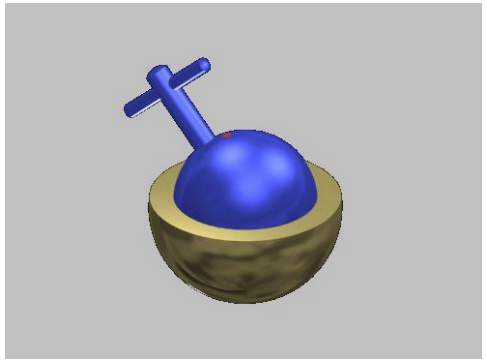
Which one usually has a larger contact pressure: surface contact or point contact?



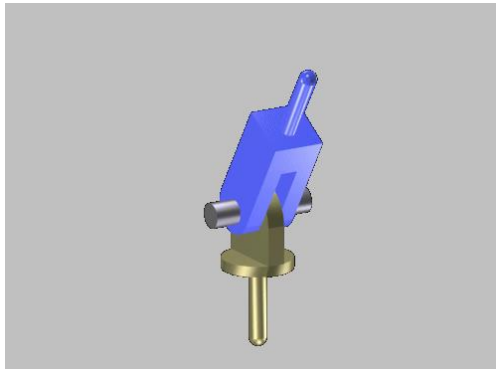
higher pair



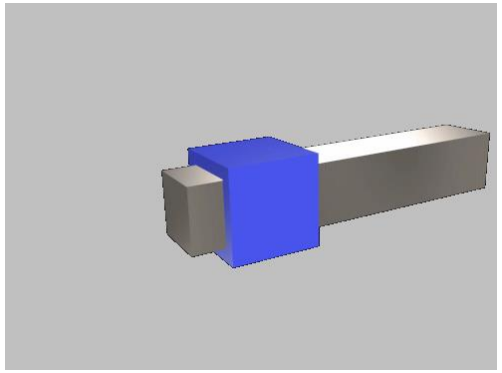
higher pair



lower pair



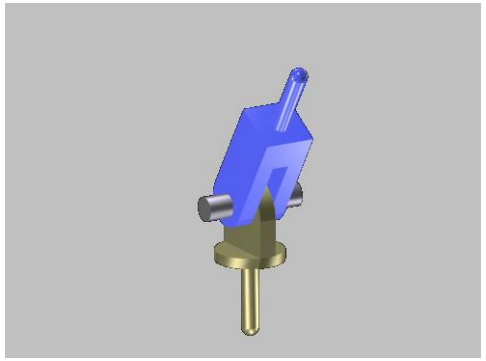
lower pair



lower pair

Imagine my hand is a rigid body.
If all its motion stays in this plane...

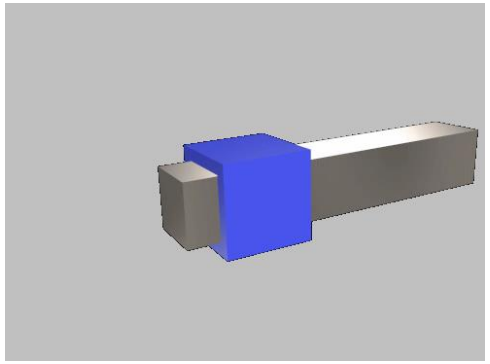
Planar or spatial?



planar

So what makes a joint planar?
Is it because the joint itself looks flat?

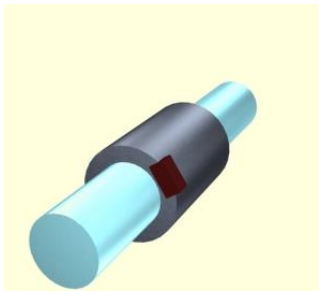
We classify a joint by its motion, not by its appearance.



planar



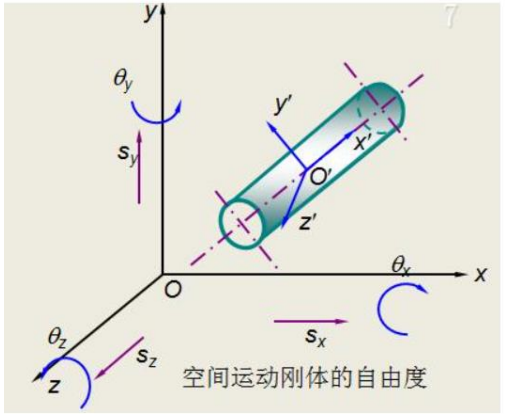
Spatial

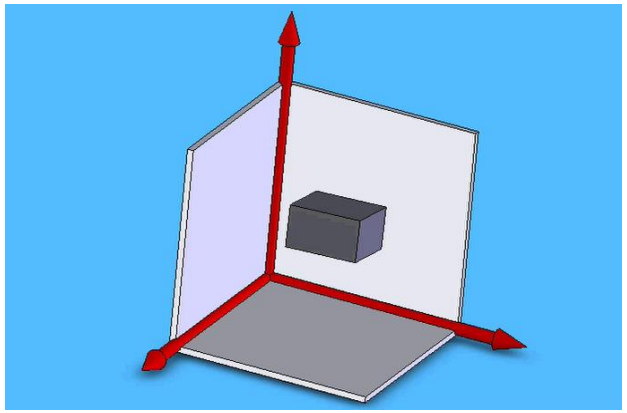


Spatial

Suppose my phone is floating in space.

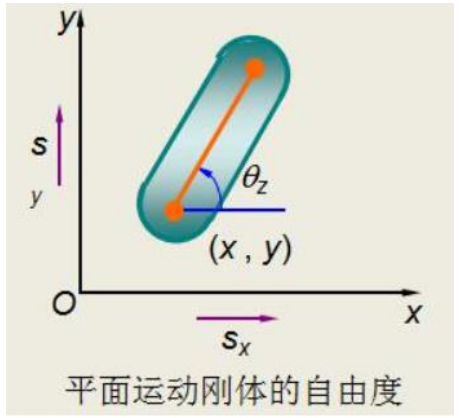
How many independent ways can it move? How many translations? How many rotations?

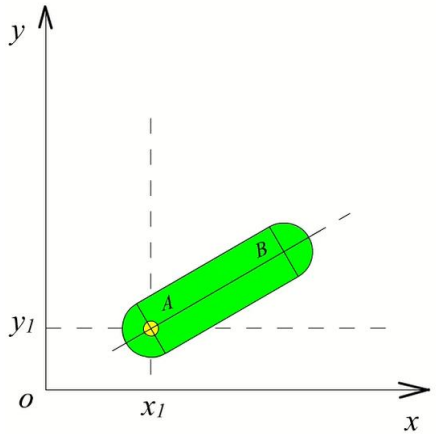




Now I put it on a table and force it to stay in this plane.

How many DOFs are left?



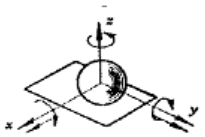
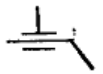
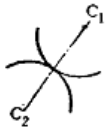
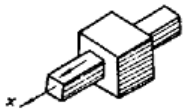
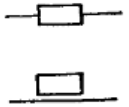
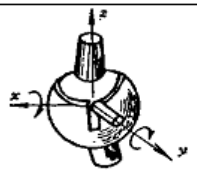


The number of independent motions of a body is called its degree of freedom, or DOF.

How many DOFs does your head have?

1-1Composition of Mechanisms and Drawing of Their Kinematic Diagrams

What happens when two links are connected by a joint?

名称	图　形	简图符号	副级	自由度	名称	图　形	简图符号	副级	自由度
球面高副			I	5	圆柱套筒副			IV	2
柱面高副			II	4	转动副			V	1
球面低副			III	3	移动副			V	1
球销副			IV	2	螺旋副			V	1

Joint class = number of constraints imposed by the joint.

Every joint takes away some freedom.
We call the removed motions **constraints**.

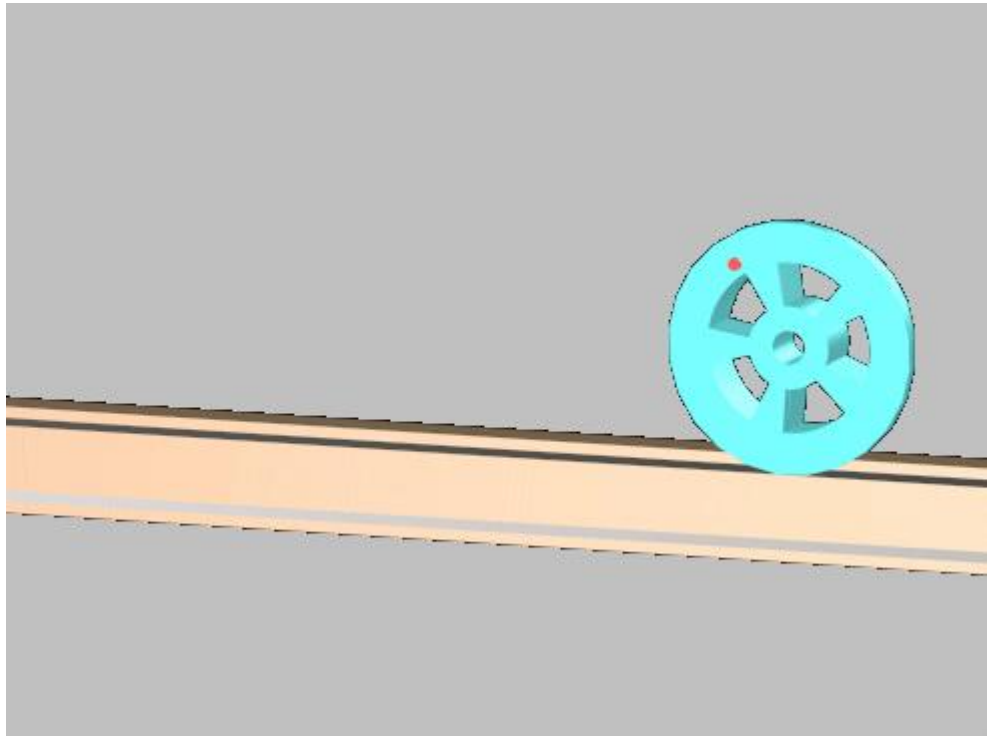
More constraints → fewer DOFs

A joint gives freedom with one hand and takes freedom away with the other.

If a free rigid body has six DOFs, and a joint leaves only one DOF, **how many independent motions has the joint removed?**

Five. This is a **Class-V joint** in this classification.

How about DOF of screw pair ?



look at the rolling wheel (take it as a planar mechanism) and answer three questions:

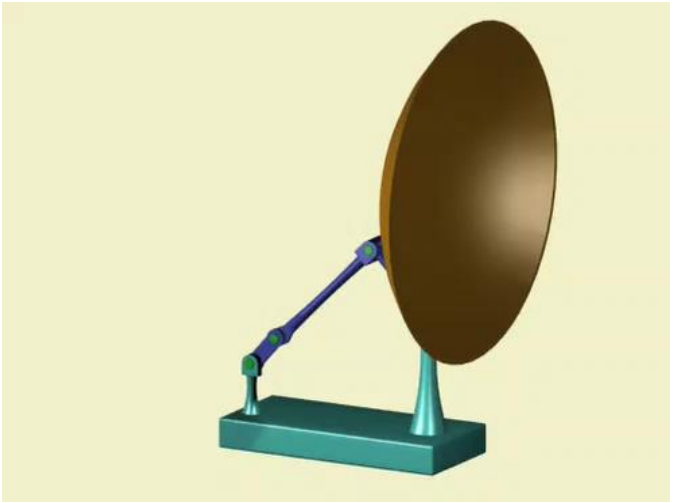
1.Higher pair or lower pair?

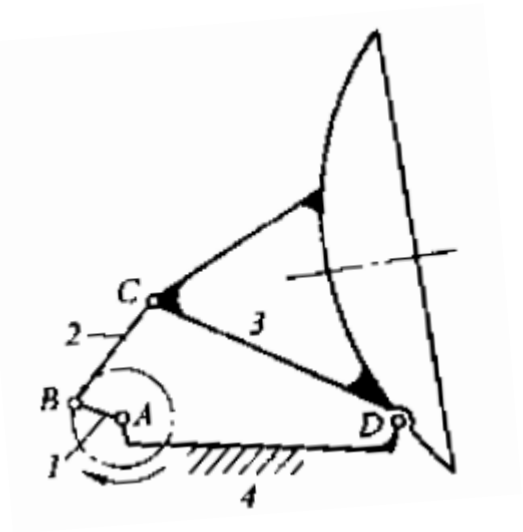
2.How many constraints does it introduce?

Exit Quiz

Machine → ? → ? → allowed motion → DOF
Links Joints

So today we learned how to "decompose" a machine.
A real machine may look complicated, but from the viewpoint of kinematics, we only need to ask:
What are the links?
How are they connected?
What motion is allowed?

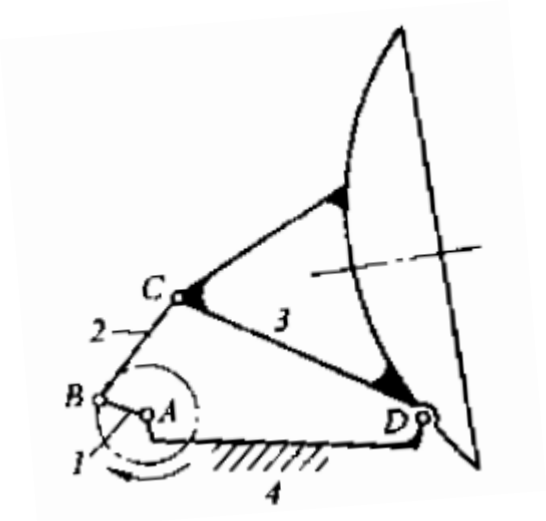


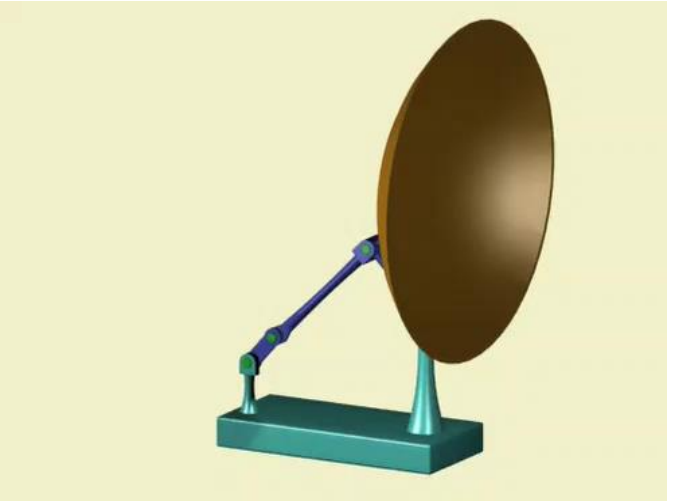


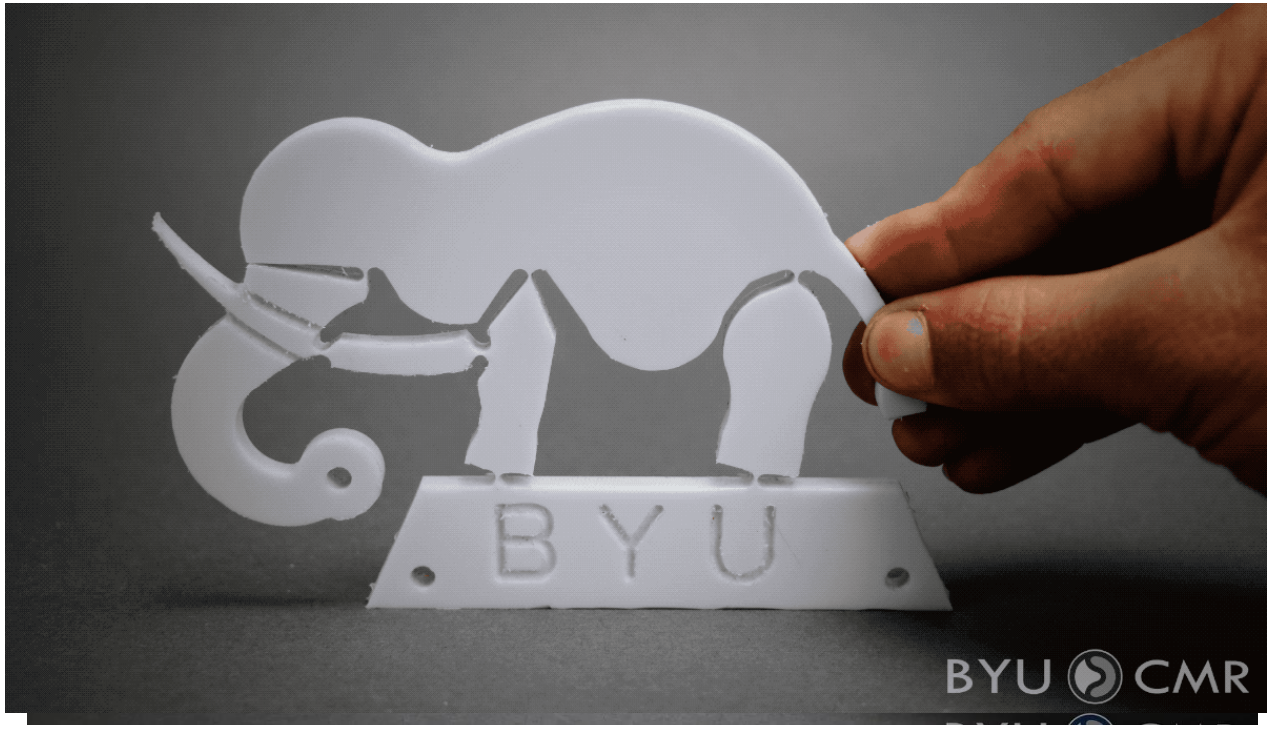
Review

Let's see whether you can now read a machine like a mechanical engineer.

How many links can you identify?
Which one is the frame?
Which is the input link?
Which is the output link?
What joints can you find?
Higher pairs or lower pairs?
Planar or spatial?





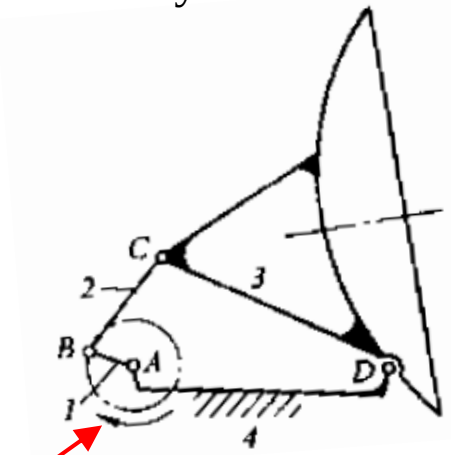


Now I have a mechanical magic trick for you.

I pull the elephant's tail to the right.

What will happen to the center of the trunk?

A. Move left
B. Move right
C. Move up
D. Move down
E. Stay almost at the same position





Radar Mechanism

Drawing of kinematic diagrams of mechanisms

1. Understand its definition and significance

2. Master its drawing method

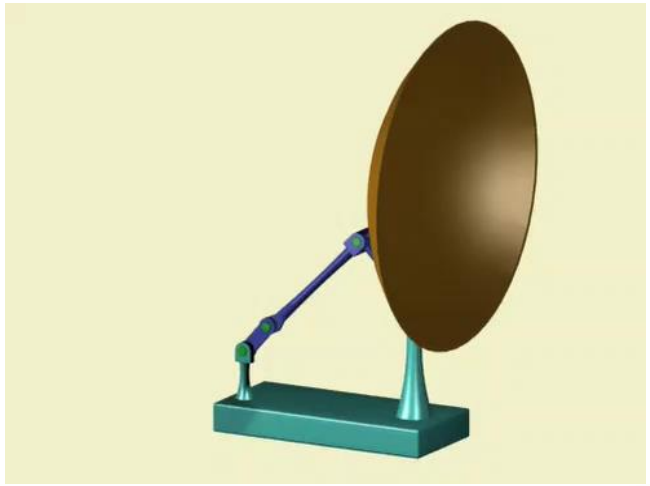
To analyze the motion of this radar, do we need its color?

Do we need the exact shape of the radar dish?

Do we need the thickness of every rod?

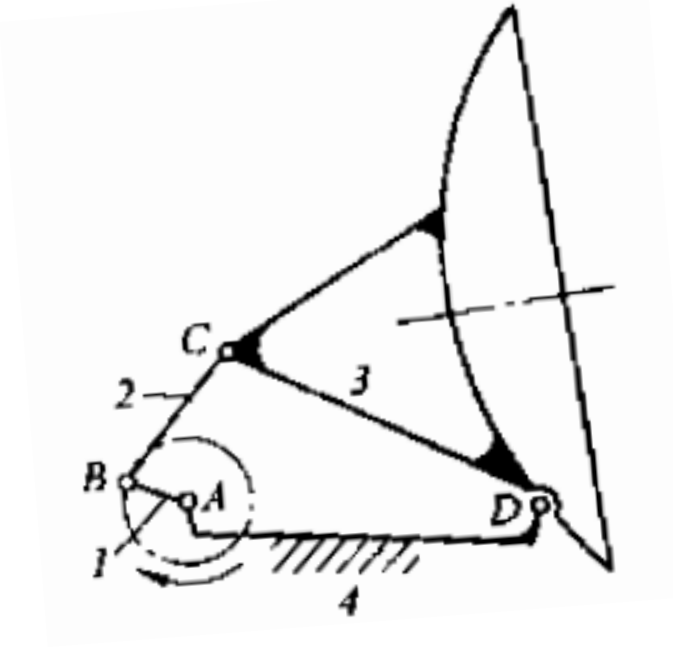
Do we need the locations of the joints?

Do we need to know how the links are connected?



A **kinematic diagram** keeps the information that determines motion and removes the information that does not.

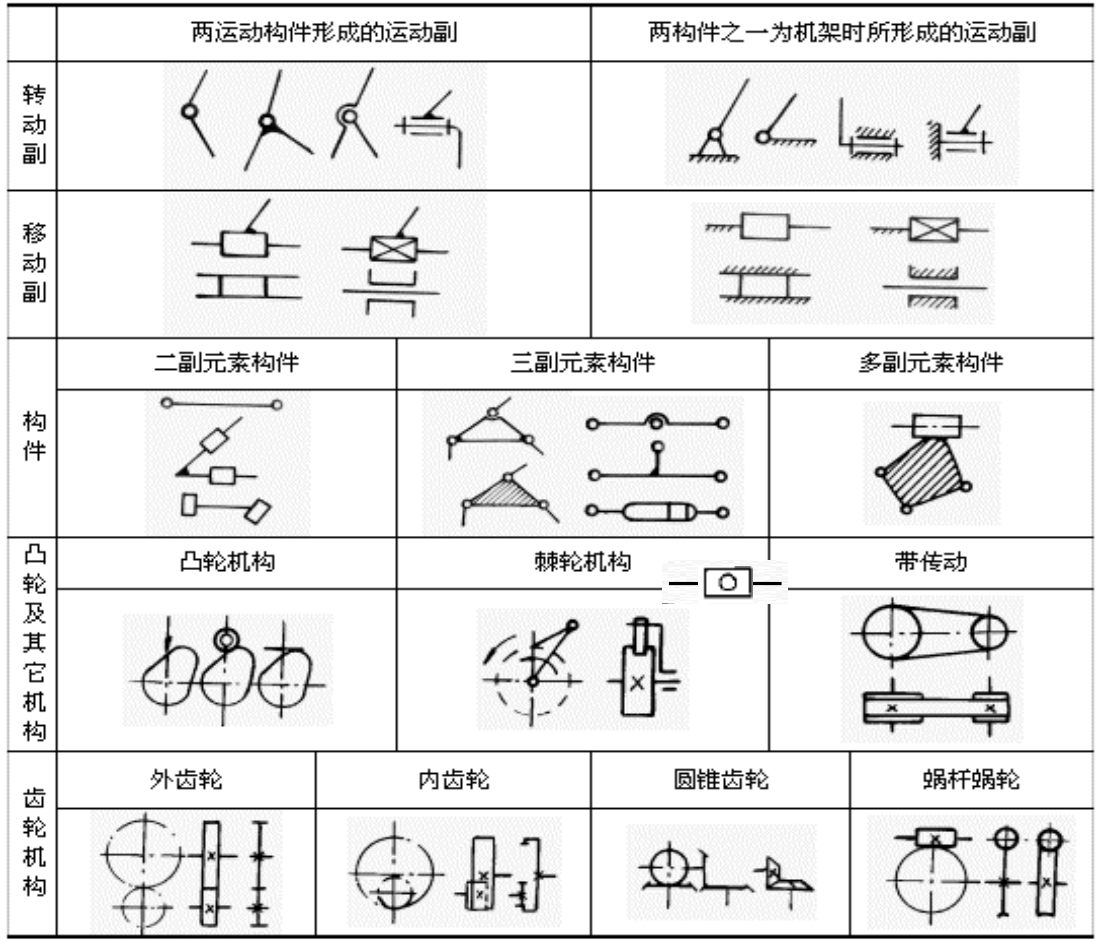
A kinematic diagram is the “skeleton” of a machine, where we use simple symbols to draw the links and joints, making it easy for expression of motion transformation.



Kinematic diagram: topology + relative geometry.

Kinematic schematic: topology is important; scale is not. (for concept design)

You do NOT need to memorize the table But You need to how to use it.



- revolute joint
- prismatic joint
- frame
- binary link
- ternary link

Question 1
What moves together?

→ Identify all links. Use Arabic numerals (1,2,3…) to label links

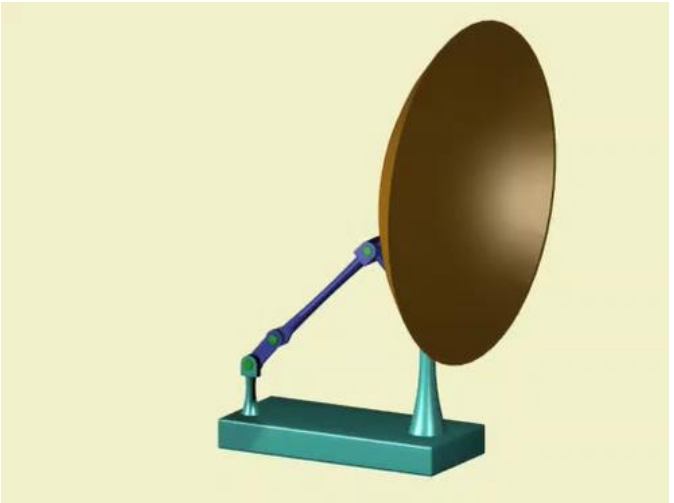
Question 2
How are they connected?
→ Identify the joints. Use Capital Letters (A, B, C…) to label R joints

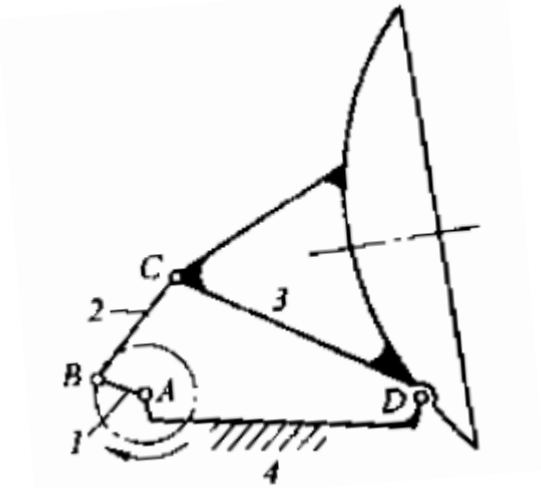
Question 3
From which direction should we observe the mechanism?
→ Select the projection plane.

Question 4
Where are the joints relative to one another?
→ Determine positions, scale, and draw.

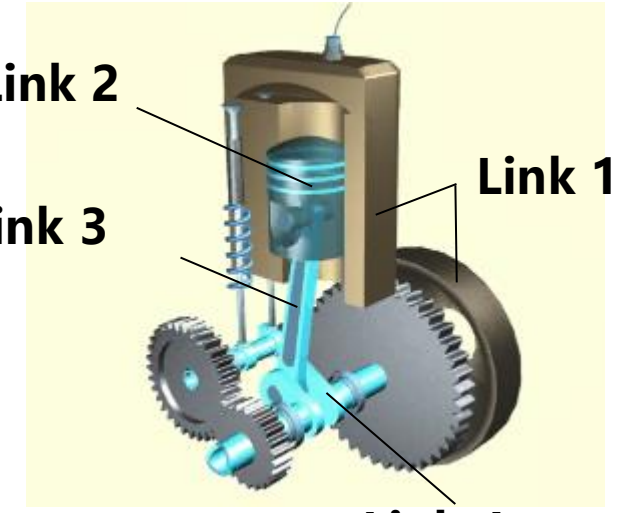
Which link drives the mechanism?
→ Mark input motion

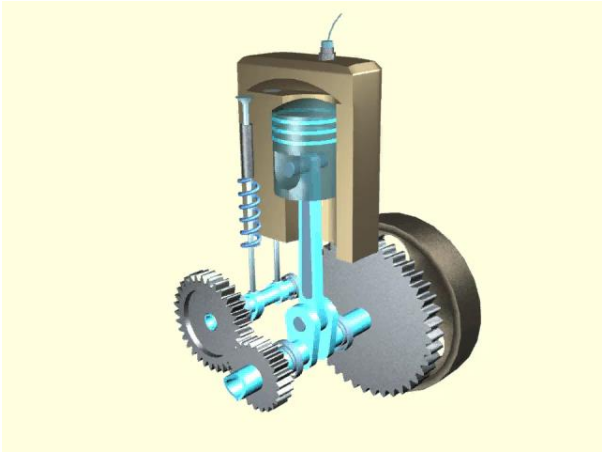
雷达摇摆机构





Combustion engine





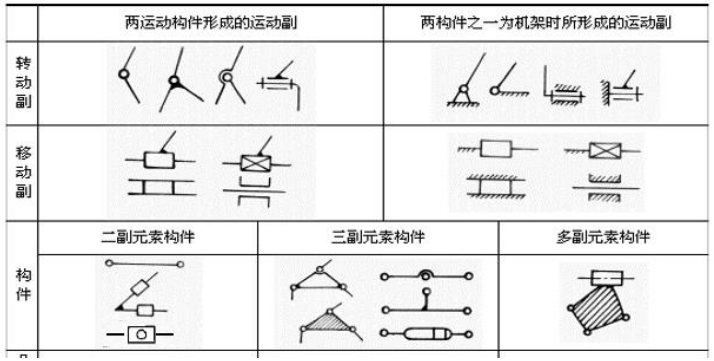
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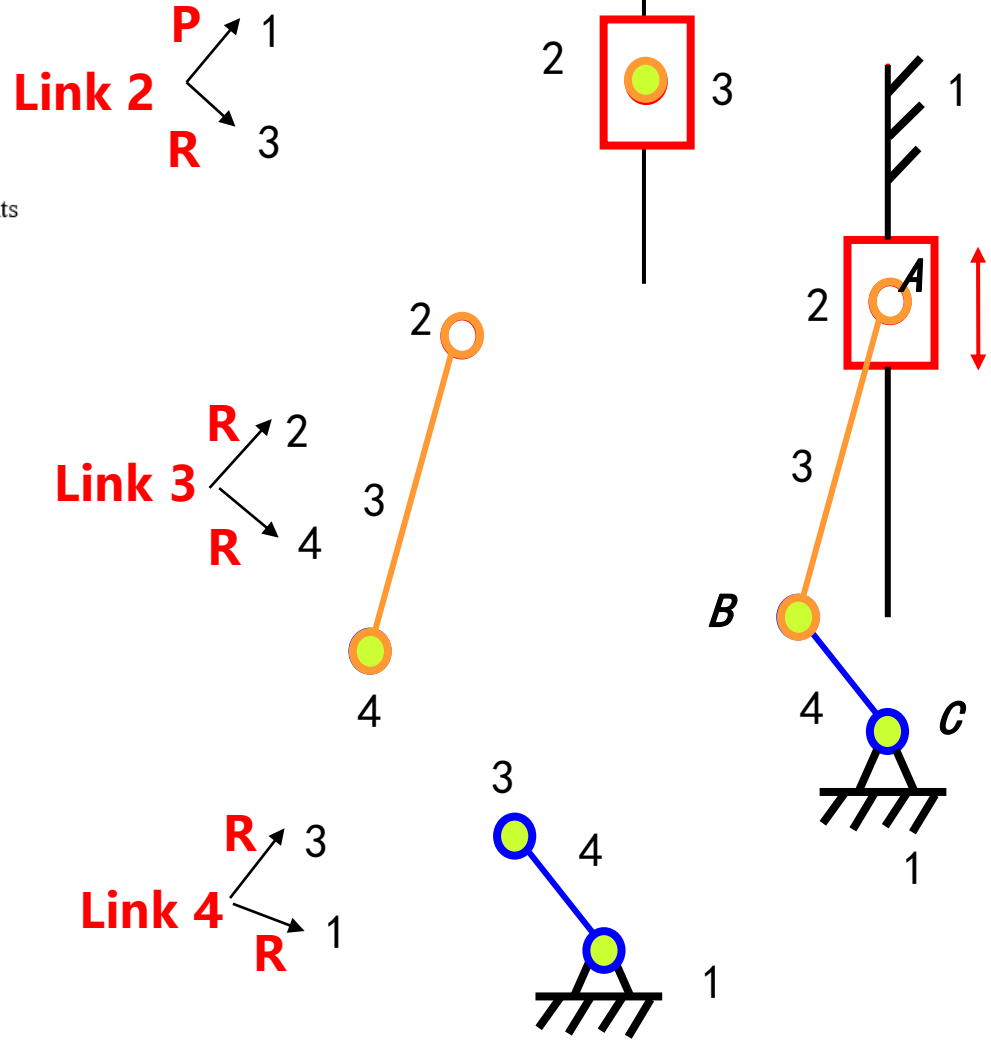
Question 2
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→ Identify the joints. Use Capital Letters (A, B, C…) to label R joints

Question 3
From which direction should we observe the mechanism?
→ Select the projection plane.

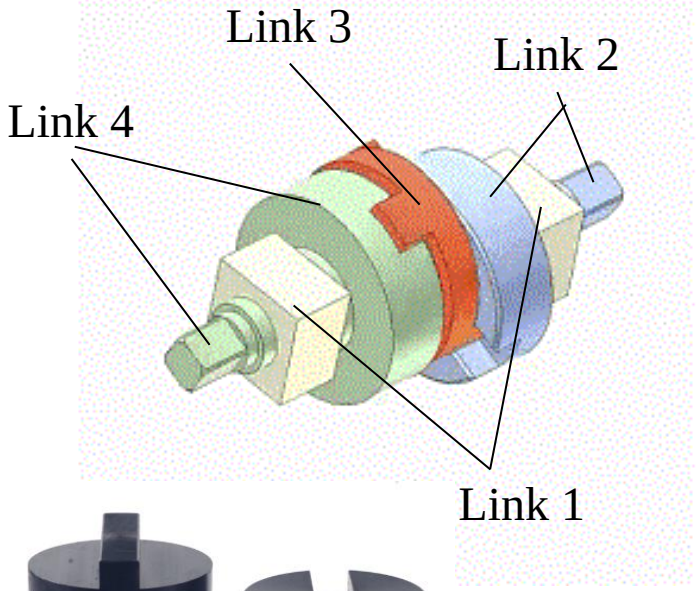
Question 4
Where are the joints relative to one another?
→ Determine positions, scale, and draw.

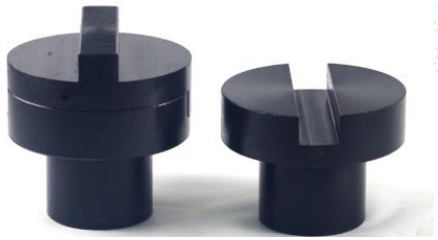
Which link drives the mechanism?
→ Mark input motion





1-1Composition of Mechanisms and Drawing of Their Kinematic Diagrams





Oldham coupling

Question 1
What moves together?

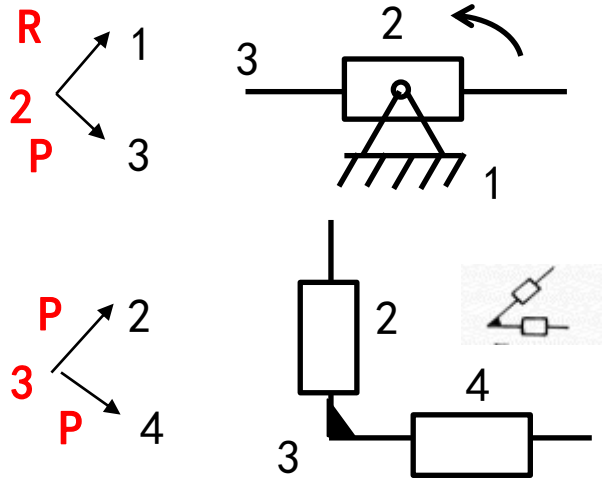
→ Identify all links. Use Arabic numerals (1,2,3…) to label links

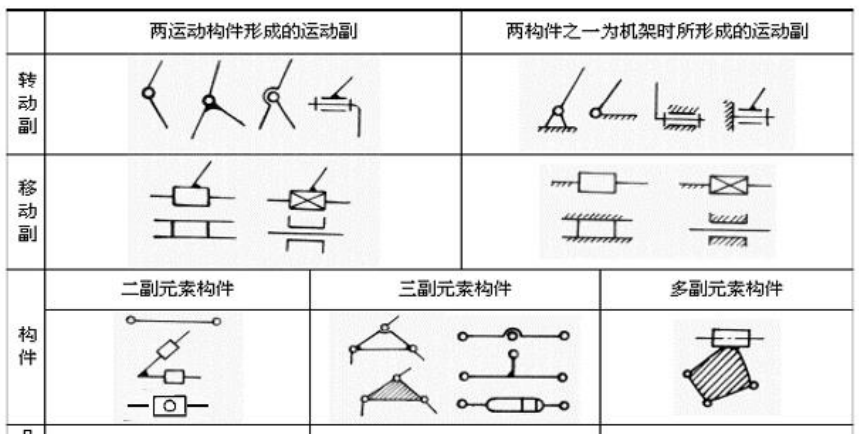
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From which direction should we observe the mechanism?
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Which link drives the mechanism?
→ Mark input motion

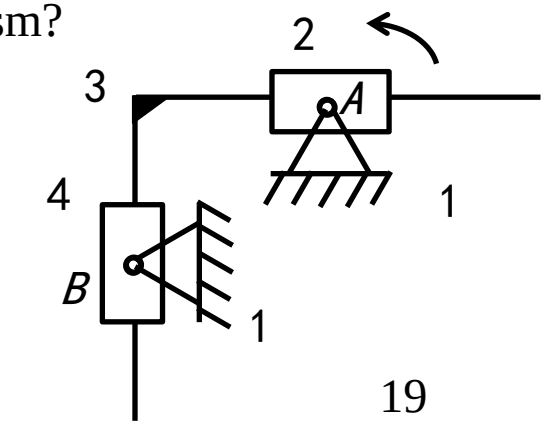


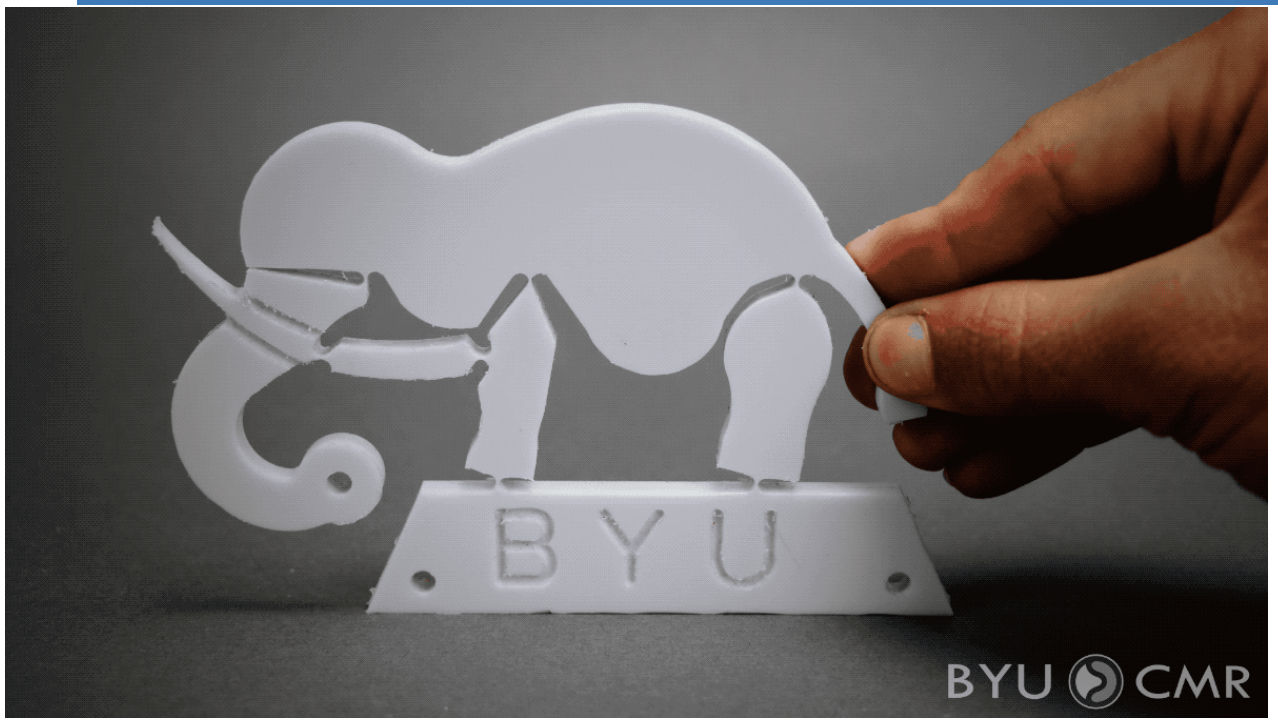


Why would anyone design such a strange mechanism?

Suppose two rotating shafts are parallel, but their axes are slightly misaligned.
Can we connect them rigidly?

Transmits rotation while tolerating some offset





Now you are engineers. Guessing is no longer enough.
Try to draw its kinematic diagram and prove it.

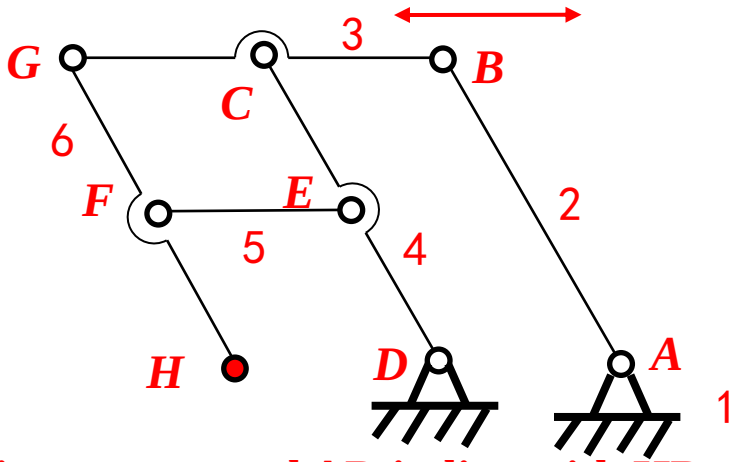
Question 1
What moves together?
→ Identify all links. Use Arabic numerals (1,2,3…) to label links

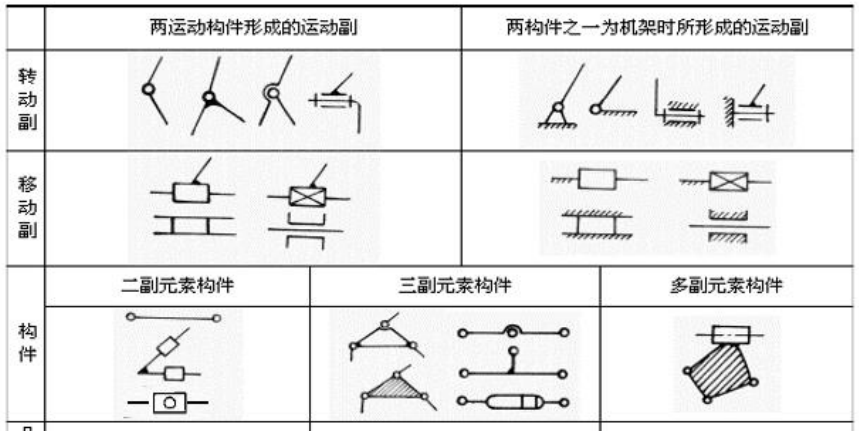
Question 2
How are they connected?
→ Identify the joints. Use Capital Letters (A, B, C…) to label R joints

Question 3
From which direction should we observe the mechanism?
→ Select the projection plane.

Question 4
Where are the joints relative to one another?
→ Determine positions, scale, and draw.

Which link drives the mechanism?
→ Mark input motion



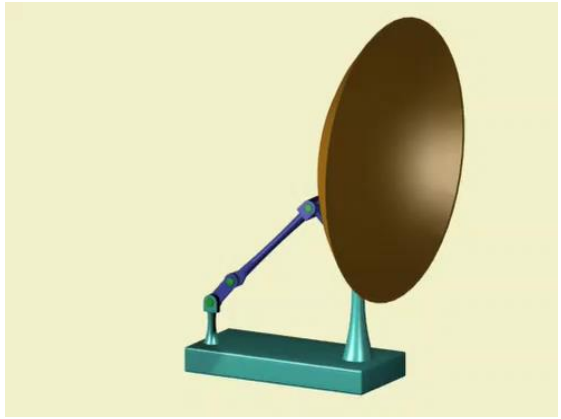


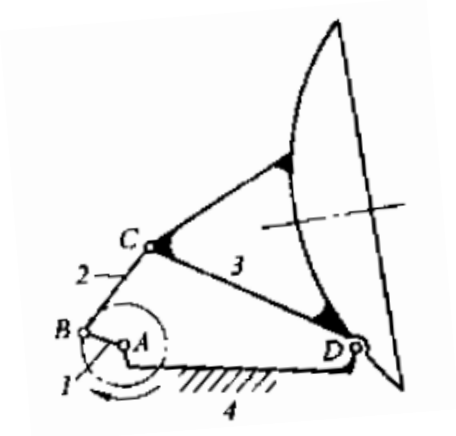
How to prove?

***H* is fixed point← *HD* is constant and AD in line with HD.**

$FH /\!/ = ED \rightarrow$ HD $/\!/ =$ FE is constant

$HD /\!/ FE /\!/ GC /\!/ DA \rightarrow$ AD in line with HD





Must a joint always be made of two separate parts?

What if relative motion comes from **elastic deformation of the material itself**?
How many parts do you see?
How many moving regions?

So where are the joints? This is a **flexure mechanism**.

What can we eliminate if we replace conventional joints with flexure hinges?

Assembly, Friction, Backlash.

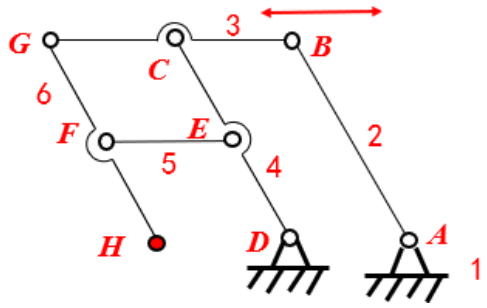
No assembly, no friction, no backlash, and potentially very high precision.

Sounds perfect. So why don't we **replace every joint** in the world with **a flexure hinge**?

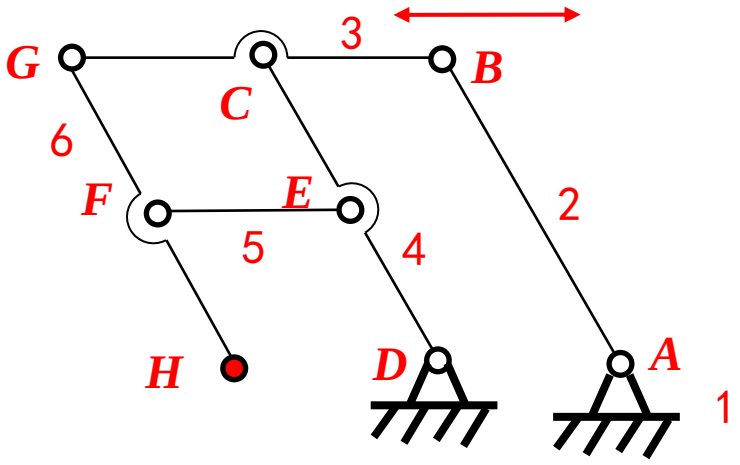
Limited motion range. Stress. Fatigue.

This is **Engineering trade-off**









Q1. What moves together?

→ Links

Q2. How are the links connected?

→ Joints

Q3. What motion is allowed and what motion is constrained?

→ DOF and constraints

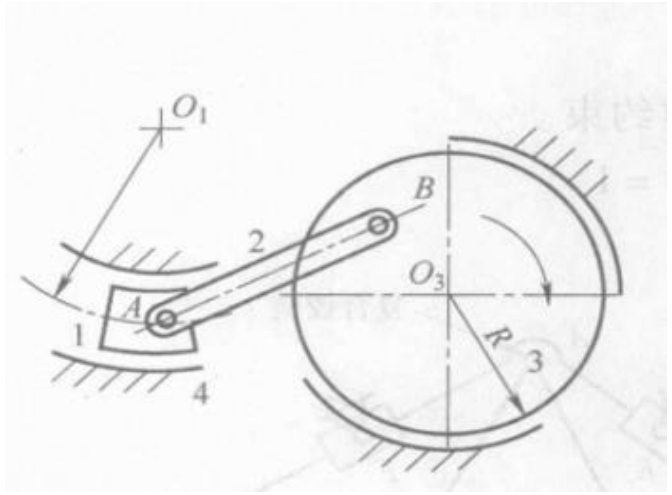
Q4. What information actually determines the motion?

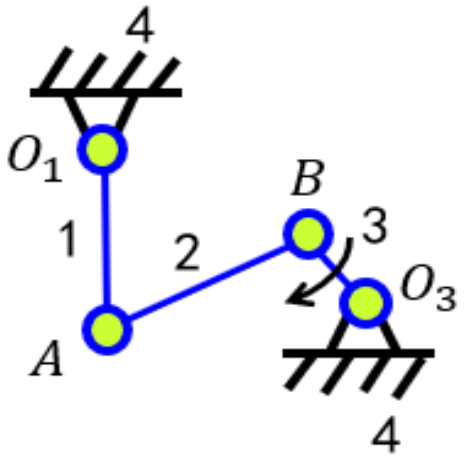
→ Kinematic diagram

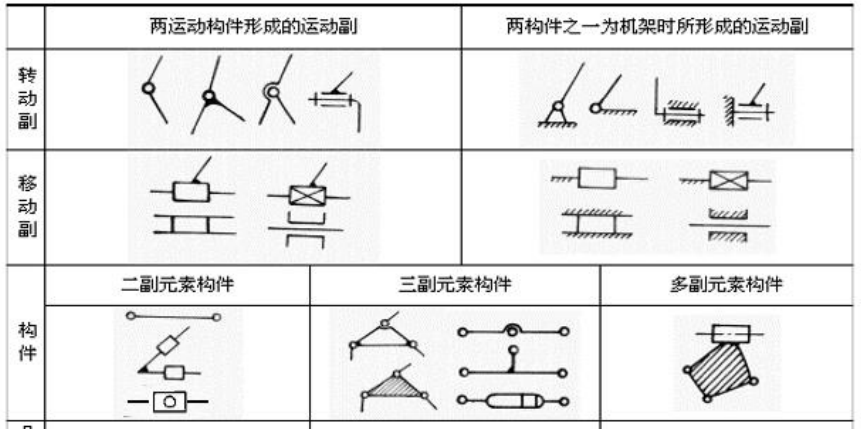
Q5. Can we predict or explain the motion from the diagram?

→ Mechanism analysis

Exit Quiz：draw its kinematic diagram







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